Akoizo Original Practice - 2027 Rotation Form 2 - Answer key

Division B | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Parallel resistors share…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. Voltage

B. Always current

C. Always power

D. Always resistance

Correct: A

2. (2 points) XOR(1,1) equals…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. Undefined

B. 0

C. 1

D. 2

Correct: B

3. (2 points) A voltmeter connects…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. Only instead of every wire

B. Only to ground alone

C. Across two nodes

D. As an intentional source short

Correct: C

4. (2 points) Two 100 Ω resistors in series give…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. 200 Ω

B. 50 Ω

C. 100 Ω

D. 10000 Ω

Correct: A

5. (2 points) NOT(A AND B) equals…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. A AND B

B. (NOT A) OR (NOT B)

C. A OR B

D. (NOT A) AND B

Correct: B

6. (2 points) An ohmmeter generally requires…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. Maximum possible voltage

B. No instrument instructions

C. A de-energized suitable circuit

D. A live mains circuit

Correct: C

7. (2 points) An open switch can have…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. Voltage with zero current

B. Infinite real power

C. Always zero voltage

D. Consumed charge

Correct: A

8. (2 points) A multiplexer…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. Measures resistance

B. Selects an input

C. Always adds inputs

D. Stores a clocked bit by definition

Correct: B

9. (2 points) Divider inference needs…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. Zero source voltage

B. No known component

C. The same series current

D. Equal resistor values

Correct: C

10. (4 points) Why trace nodes first?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

answer: Topology follows connections, not visual placement. (4 points).

11. (2 points) A 8 V ideal battery drives a 4 ohm resistor. Find current.

answer: 2 A (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , A.

12. (2 points) A series pair has resistances 4 and 6 ohms. Find equivalent resistance.

answer: 10 ohms (2 points).

Numerical target 10; absolute tolerance 0.02. Accepted units: , ohms.

13. (2 points) Two identical 8 ohm resistors are placed in parallel. Find equivalent resistance.

answer: 4 ohms (2 points).

Numerical target 4; absolute tolerance 0.008. Accepted units: , ohms.

14. (2 points) An ideal resistor carries 2 A at 8 V. Find power.

answer: 16 W (2 points).

Numerical target 16; absolute tolerance 0.032. Accepted units: , W.

15. (2 points) [State/National] A node has incoming currents 5 A and 2 A and one outgoing current. Find the outgoing current.

answer: 7 A (2 points).

Numerical target 7; absolute tolerance 0.014. Accepted units: , A.

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

answer: It measures potential difference across a component and has very high input resistance; placing it in series would alter or interrupt the intended current path. (2 points).

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

answer: 0; an AND output is 1 only when both inputs are 1. (2 points).

18. (2 points) Describe a diagnostic use of a ground-fault circuit interrupter without claiming it replaces every protection device.

answer: It detects a current imbalance between outgoing and returning paths and interrupts the circuit; overcurrent and other hazards may require additional protection. (2 points).

19. (2 points) [State/National] A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

answer: 5 V; Kirchhoff voltage law requires the algebraic sum of potential changes around the closed loop to be zero. (2 points).

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

answer: Brightness depends on operating current; matched LEDs with suitable branch current limiting and comparable currents can produce comparable brightness. Do not connect them without appropriate current control. (2 points).